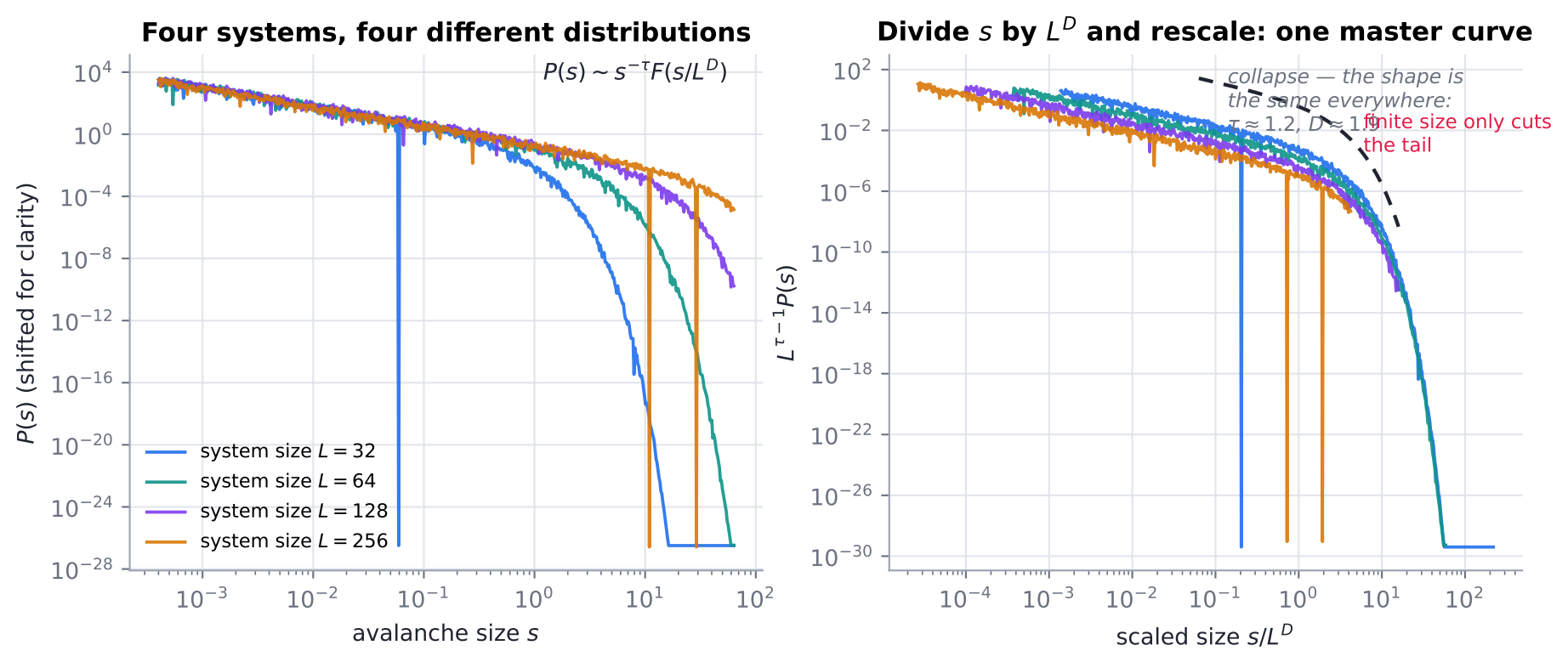

Four systems, four different distributions
$P(s) \sim s^{-\tau} F(s/L^D)$
$P(s)$ (shifted for clarity)
avalanche size s
system size $L = 32$
system size $L = 64$
system size $L = 128$
system size $L = 256$
Divide s by L^D and rescale: one master curve
collapse — the shape is
the same everywhere:
$\tau \approx 1.2$, $D \approx 1.9$
finite size only cuts
the tail
$L^{\tau-1} P(s)$
scaled size s/L^D